

PRACTICAL : 9

AIM : Cleaning and Structuring Data for Analysis

- a) Identify and handle missing or inconsistent data
- b) Convert wide data into tall format
- c) Perform union of multiple files
- d) Execute cross-database joins
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets
- g) Apply advanced data cleaning techniques

Dataset: Custom Designed Food_Delivery Dataset [csv files]

Food_Delivery_Orders.csv

Order_ID, Customer_Name, Restaurant_Code, Order_Value, Delivery_Time, Food_Rating, Delivery_Rating, Discount, Customer_Satisfaction

O101	Aarav Sharma	RES-01	450,32	4.5,4.2	10%,85 %
O102	Priya Verma	RES-02	28,4.0	4.5,15%,90 %	
O103	Rohan Gupta	RES-01	620,45	3.8,5%,78 %	
O104	Sneha Patel	RES-03	380,25	4.8,4.7,20%,95 %	
O105	Aditya Rao	RES-04	750,,3.5,3.2	25%,65 %	
O106	Ananya Singh	RES-01	520,35	4.6,4.4,10%,88 %	
O107	Rahul Nair	RES-02	40,3.8,3.9	15%,72 %	
O108	Kavya Reddy	RES-03	890,22	4.9,4.8,30%,98 %	
O109	Vikram Joshi	RES-01	340,50	3.2,3.0,5%,80 %	
O110	Pooja Iyer	RES-02	580,30	4.3,4.1,20%,91 %s	

Food_Delivery_Orders_2.csv

Order_ID, Customer_Name, Restaurant_Code, Order_Value, Delivery_Time, Food_Rating, Delivery_Rating, Discount, Customer_Satisfaction

O201	Aarav Sharma	RES-01	520,30	4.6,4.4,12%,88 %
O202	Priya Verma	RES-02	680,35	4.2,4.3,18%,92 %
O203	Rohan Gupta	RES-01	590,40	4.3,4.0,10%,82 %
O204	Sneha Patel	RES-03	420,24	4.9,4.8,25%,96 %

T120 DHEERAJ SINGH

Data Visualization with Power BI and Tableau

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

O205,Aditya Rao,RES-04,810,38,3.8,3.7,30%,70 %
O206,Ananya Singh,RES-01,560,32,4.7,4.5,15%,90 %
O207,Rahul Nair,RES-02,490,42,4.0,4.1,20%,75 %
O208,Kavya Reddy,RES-03,920,20,5.0,4.9,35%,99 %
O209,Vikram Joshi,RES-01,390,48,3.5,3.4,8%,83 %
O210,Pooja Iyer,RES-02,610,29,4.4,4.2,22%,94 %

Restaurant_Master.csv

Restaurant_Code,Restaurant_Name,Cuisine,City,Target_Rating,Daily_Order_Capacity
RES-01,Spice Garden,Indian,Mumbai,4.3,150
RES-02,Urban Tadka,North Indian,Pune,4.1,120
RES-03,Food Street,Multi-Cuisine,Mumbai,4.5,100
RES-04,The Food Hub,Continental,Delhi,4.0,80

Customer_Activity.csv

Customer_Name,Loyalty_Program,Orders_Completed,Offers_Used,App_Rating
Aarav Sharma,Gold,12,5,4.6
Priya Verma,Silver,8,4,4.3
Rohan Gupta,Gold,10,3,4.1
Sneha Patel,Platinum,15,7,4.8
Aditya Rao,Bronze,5,2,3.7
Ananya Singh,Gold,13,5,4.7
Rahul Nair,Silver,7,3,4.0
Kavya Reddy,Platinum,18,8,4.9
Vikram Joshi,Bronze,6,2,3.6
Pooja Iyer,Gold,11,5,4.5

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A: Identify and Handle Missing or Inconsistent Data

Tableau - Book1

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Advertising ...nd Sales.csv
Agriculture Production.csv
Category_Icons.csv
Customer_Activity.csv
Data Set- In...ist_2014.csv
Department_Master.csv
fifa18_clean.csv
Food_Delivery_Orders.csv
Food_Delive...rders_2.csv
INDIA_RETAIL_DATA.csv
indian_start...funding.csv
List of Orders.csv

New Union
New Table E

Go to Worksheet

Data Source

Sheet 1

Food_Delivery_Orders

Food_Delivery_Orders.csv

9 fields 10 rows

10 rows

Name Food_Delivery_Orders.csv

Description
No description available.

Fields

Type	Field Name	Physical Ta...	Rem...
Abc	Order ID	Food_Deliver...	Order...
Abc	Customer Name	Food_Deliver...	Custo...

Order ID	Customer Name	Restaurant Code	Order Value	Delivery
O101	Aarav Sharma	RES-01	450	
O102	Priya Verma	RES-02	null	
O103	Rohan Gupta	RES-01	620	
O104	Sneha Patel	RES-03	380	
O105	Aaritva Rao	RFS-04	750	

Tableau - T120 DHEERAJ SINGH 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Data Analytics

Food_Delivery_Orders

Search

Tables

Abc Customer Name
Abc Customer Satisfaction
Abc Order ID
Abc Restaurant Code
Abc Measure Names

Delivery Rating
Delivery Time
Discount
Food Rating
Order Value
Food_Delivery_Orders.csv...
Measure Values

Columns

Rows

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Discount_Clean

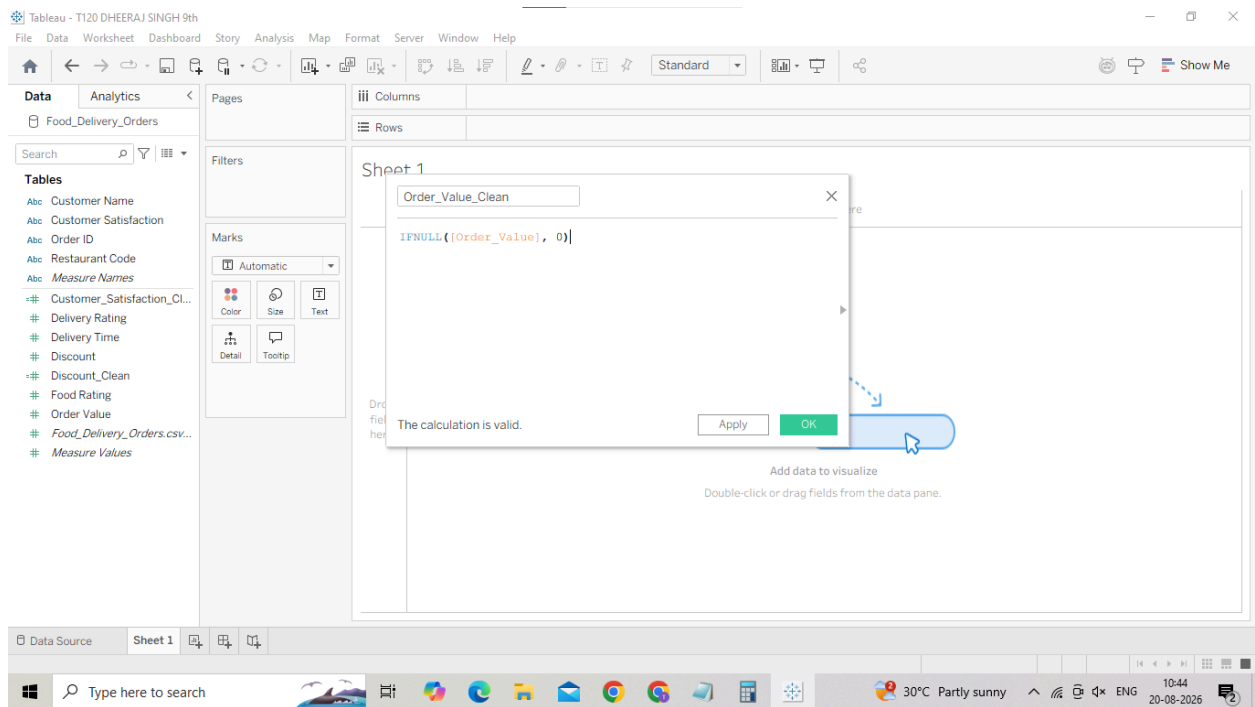
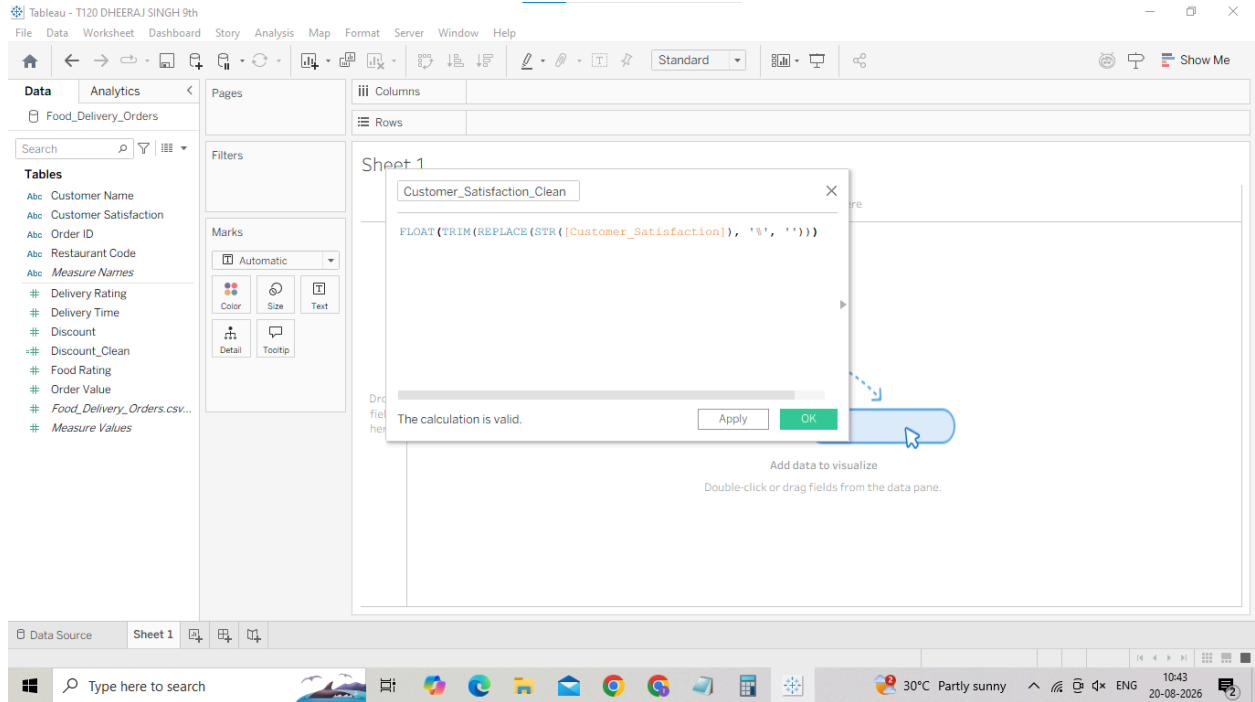
Float (Trim (Replace (Str ([Discount]), '%', '')))

The calculation is valid.

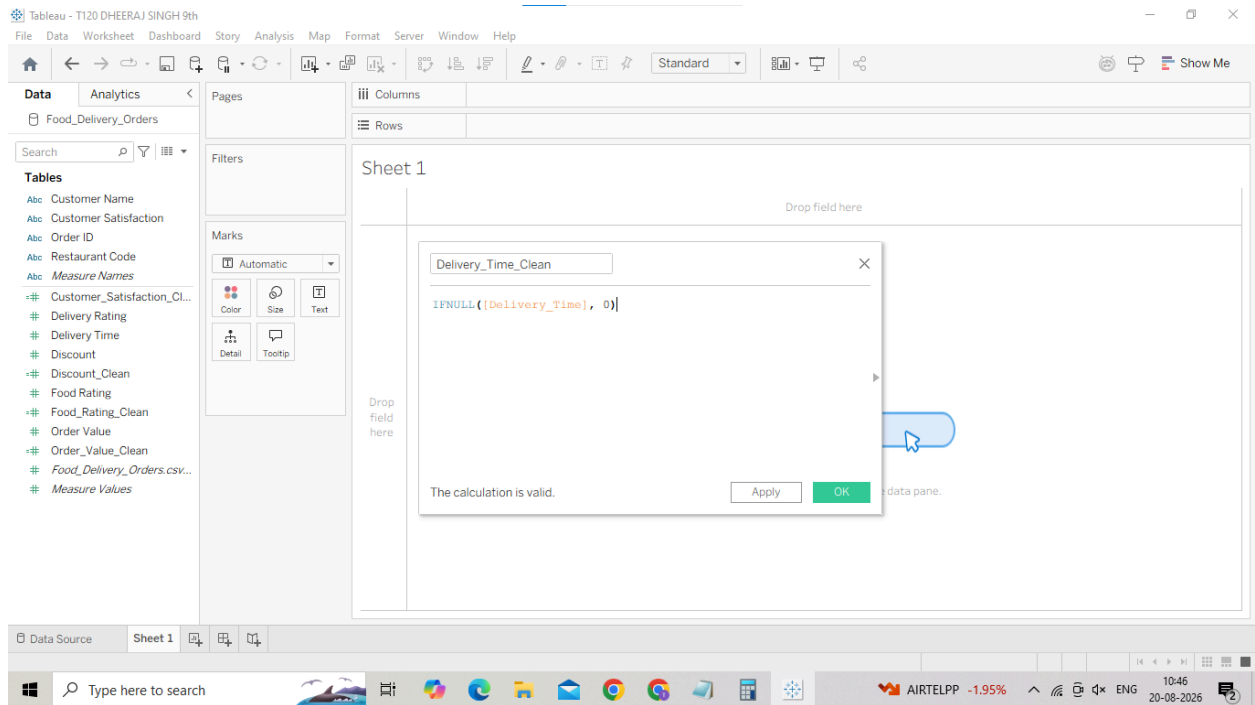
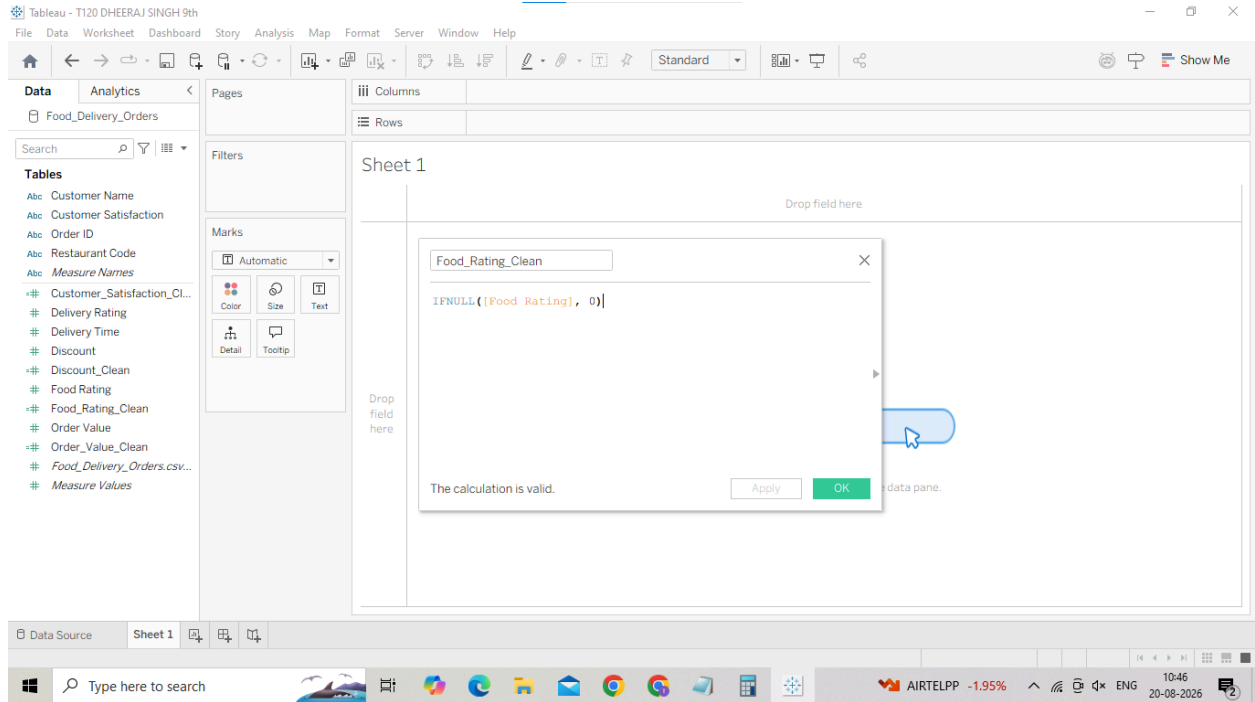
Apply OK

Add data to visualize
Double-click or drag fields from the data pane.

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B: Convert Wide Data into Tall Format

Tableau - T120 DHEERAJ SINGH 9th

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Advertising_and Sales.csv
- Agriculture Production.csv
- Category_Icons.csv
- Customer_Activity.csv
- Data Set- In...ist_2014.csv
- Department_Master.csv
- fifa18_clean.csv
- Food_Delivery_Orders.csv
- Food_Delive...rders_2.csv
- INDIA_RETAIL_DATA.csv
- indian_start...funding.csv
- List of Orders.csv

New Union

New Table Extension

Food_Delivery_Orders

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Delivery_Orders.csv

Food_Delivery_Orders.csv 14 fields 10 rows

10 rows

Name Food_Delivery_Orders.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Order ID	Food_Delivery_...	Order...
Abc	Customer Name	Food_Delivery_...	Custo...

Order ID	Customer Name	Restaurant Code	Order Value	Deliver
O101	Aarav Sharma	RES-01	450	
O102	Priya Verma	RES-02	null	
O103	Rohan Gupta	RES-01	620	
O104	Sneha Patel	RES-03	380	
O105	Aarav Rao	RES-04	750	

Data Source Sheet 1

Type here to search

They found cold resc...

10:49 20-08-2026

Tableau - T120 DHEERAJ SINGH 9th

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Advertising_and Sales.csv
- Agriculture Production.csv
- Category_Icons.csv
- Customer_Activity.csv
- Data Set- In...ist_2014.csv
- Department_Master.csv
- fifa18_clean.csv
- Food_Delivery_Orders.csv
- Food_Delive...rders_2.csv
- INDIA_RETAIL_DATA.csv
- indian_start...funding.csv
- List of Orders.csv

New Union

New Table Extension

Food_Delivery_Orders

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Delivery_Orders.csv

Relate tables
Drag tables near Food_Delivery_Orders.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Food_Delivery_Orders.csv 14 fields 10 rows

10 rows

Name Food_Delivery_Orders.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Restaurant Code	Food_Delivery_...	Resto...
Abc	Order Value	Food_Delivery_...	Order...
Abc	Delivery Time	Food_Delivery_...	Delive...
Abc	Food Rating	Food_Delivery_...	Food...
Abc	Delivery Rating	Food_Delivery_...	Delive...

Order ID	Customer Name	Restaurant Code	Order Value	Delivery Time	Food Rating	Delivery Rating	Discount
O101	Aarav Sharma	RES-01	450	32	4.50000	4.20000	0.1
O102	Priya Verma	RES-02	null	28	4.00000	4.50000	0.1
O103	Rohan Gupta	RES-01	620	45	null	3.80000	0.0
O104	Sneha Patel	RES-03	380	25	4.80000	4.70000	0.2
O105	Aarav Rao	RES-04	750	null	3.50000	3.20000	0.1
O106	Ananya Singh	RES-01	520	35	4.60000	4.40000	0.1
O107	Rahul Nair	RES-02	null	40	3.80000	3.90000	0.1
O108	Kanya Reddy	RES-03	890	22	4.90000	4.80000	0.2

Data Source Sheet 1

Type here to search

10:50 20-08-2026

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Tableau - T120 DHEERAJ SINGH 9th

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Advertising ...nd Sales.csv
Agriculture Production.csv
Category_Icons.csv
Customer_Activity.csv
Data Set- In...ist_2014.csv
Department_Master.csv
fifa18_clean.csv
Food_Delivery_Orders.csv
Food_Delive...rders_2.csv
INDIA_RETAIL_DATA.csv
indian_start...funding.csv
List of Orders.csv

New Union

New Table Extension

Food_Delivery_Orders

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Delivery_Orders.csv

Food_Delivery_Orders.csv 12 fields 40 rows

40 rows

Name Food_Delivery_Orders.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Metric	Pivot	Pivot...
#	Value	Pivot	Pivot...
+ #	Discount_Clean	Calculation	Calcul...

Abc	Abc	Abc	#	Abc
Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv	Food_Delivery_Orders.csv
Order ID	Customer Name	Restaurant Code	Discount	Customer Satisfaction
O101	Aarav Sharma	RES-01	0.100000	85 %
O101	Aarav Sharma	RES-01	0.100000	85 %
O101	Aarav Sharma	RES-01	0.100000	85 %
O101	Aarav Sharma	RES-01	0.100000	85 %
O102	Priya Verma	RES-02	0.150000	90 %
O102	Priva Verma	RES-02	0.150000	90 %

Tableau - T120 DHEERAJ SINGH 9th

File Data Server Window Help

Connections

Food_Delivery_Orders
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Advertising ...nd Sales.csv
Agriculture Production.csv
Category_Icons.csv
Customer_Activity.csv
Data Set- In...ist_2014.csv
Department_Master.csv
fifa18_clean.csv
Food_Delivery_Orders.csv
Food_Delive...rders_2.csv
INDIA_RETAIL_DATA.csv
indian_start...funding.csv
List of Orders.csv

New Union

New Table Extension

Food_Delivery_Orders

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Delivery_Orders.csv

Food_Delivery_Orders.csv 12 fields 40 rows

40 rows

Name Food_Delivery_Orders.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Metric	Pivot	Pivot...
#	Value	Pivot	Pivot...
+ #	Discount_Clean	Calculation	Calcul...

Abc	Abc	Abc	#	+ #	+ #
Food_Delivery_Orders.csv	Pivot	Pivot	Pivot	Calculation	Calculation
Customer Satisfaction	Metric	Value	Pivot.Pivot Field Values	Discount_Clean	Customer_Sat
00	85 %	Delivery Rating	4.200	0.100000	
00	85 %	Delivery Time	32.000	0.100000	
00	85 %	Food Rating	4.500	0.100000	
00	85 %	Order Value	450.000	0.100000	
00	90 %	Delivery Rating	4.500	0.150000	
00	90 %	Delivery Time	28.000	0.150000	

C: Perform Union of Multiple Files

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Advertising...nd Sales.csv
Agriculture Production.csv
Category_Icons.csv
Customer_Activity.csv
Data Set- In...ist_2014.csv
Department_Master.csv
fifa18_clean.csv
Food_Delivery_Orders.csv
Food_Delive...rders_2.csv
INDIA_RETAIL_DATA.csv
indian_start...funding.csv
List of Orders.csv

New Union
New Table Extension

Patient_Records

Open...
Rename
Suggestions...
Remove
☒ Field names are in first row
Generate field names automatically
Text File Properties...
Convert to Union...

Patient_Records.csv 11 fields 10 rows

Name Patient_Records.csv

Description
No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Patient ID	Patient_Recor...	Patien...
Abc	Patient Name	Patient_Recor...	Patien...

Patient ID	Patient Name	Department	Age	Gender	Diagn...
P201	Rahul Mehta	Cardiology	54	Male	Hype
P202	Anita Sharma	Orthopedics	42	Female	Fract
P203	Vikram Singh	Neurology	61	Male	Migr
P204	Priya Nair	Pediatrics	12	Female	Fever
P205	Ariun Patel	Cardiology	67	Male	Hear

Data Source Sheet 1

Type here to search

28°C Mostly cloudy 08:33 27-08-2026

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Advertising...nd Sales.csv
Agriculture Production.csv
Category_Icons.csv
Customer_Activity.csv
Data Set- In...ist_2014.csv
Department_Master.csv
fifa18_clean.csv
Food_Delivery_Orders.csv
Food_Delive...rders_2.csv
INDIA_RETAIL_DATA.csv
indian_start...funding.csv
List of Orders.csv

New Union
New Table Extension

Patient_Records

Drag tables here to create a data model

[Learn more](#)

Data Source Sheet 1

Type here to search

28°C Mostly cloudy 08:35 27-08-2026

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Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Advertising ...nd Sales.csv
Agriculture Production.csv
Category_Icons.csv
Customer_Activity.csv
Data Set- In...ist_2014.csv
Department_Master.csv
fifa18_clean.csv
Food_Delivery_Orders.csv
Food_Delivery_Orders_2.csv
INDIA_RETAIL_DATA.csv
indian_start...funding.csv
List of Orders.csv

New Union
New Table Extension

Food_Delivery_Orders.csv+

Specific (manual) Wildcard (automatic)

Connection: Patient_Records

Food_Delivery_Orders.csv
Food_Delivery_Orders_2.csv

Tables in union: 2

Apply OK

Name Food_Delivery_Orders.csv

Description
No description available.

Fields

Type	Field Name	Physical Ta...	Rem...
Abc	Order ID	Food_Deliver...	Order...
Abc	Customer Name	Food_Deliver...	Custo...

Food_Delivery_Orders.csv

20 rows

Abc Food_Delivery_Orders.csv	Abc Food_Delivery_Orders.csv	Abc Food_Delivery_Orders.csv+	# Food_Delivery_Orders.csv	# Food_Deliv...
Order ID	Customer Name	Restaurant Code	Order Value	Delivery
O101	Aarav Sharma	RES-01	450	
O102	Priya Verma	RES-02	null	
O103	Rohan Gupta	RES-01	620	
O104	Sneha Patel	RES-03	380	
O105	Arlitva Rao	RFS-04	750	

Data Source Sheet 1

Type here to search

28°C Mostly cloudy 08:37 27-08-2026

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Advertising ...nd Sales.csv
Agriculture Production.csv
Category_Icons.csv
Customer_Activity.csv
Data Set- In...ist_2014.csv
Department_Master.csv
fifa18_clean.csv
Food_Delivery_Orders.csv
Food_Delivery_Orders_2.csv
INDIA_RETAIL_DATA.csv
indian_start...funding.csv
List of Orders.csv

New Union
New Table Extension

Food_Delivery_Orders.csv

10 fields 20 rows

Name Food_Delivery_Orders.csv

Description
No description available.

Fields

Type	Field Name	Physical Ta...	Rem...
Abc	Order ID	Food_Deliver...	Order...
Abc	Customer Name	Food_Deliver...	Custo...

Food_Delivery_Orders.csv

20 rows

Abc Food_Delivery_Orders.csv	Abc Food_Delivery_Orders.csv	Abc Food_Delivery_Orders.csv+	# Food_Delivery_Orders.csv	# Food_Deliv...
Order ID	Customer Name	Restaurant Code	Order Value	Delivery
O101	Aarav Sharma	RES-01	450	
O102	Priya Verma	RES-02	null	
O103	Rohan Gupta	RES-01	620	
O104	Sneha Patel	RES-03	380	
O105	Arlitva Rao	RFS-04	750	

Data Source Sheet 1

Type here to search

28°C Mostly cloudy 08:37 27-08-2026

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The screenshot shows the Tableau Desktop interface. The top menu bar includes File, Data, Server, Window, and Help. The left sidebar shows the 'Connections' pane with 'Patient_Records' selected, and the 'Files' pane listing various CSV files. The main workspace displays the 'Patient_Records' data source, which is a text file. A preview of the 'Food_Delivery_Orders.csv' file is shown, displaying 10 fields and 20 rows. The preview table has columns: Delivery Rating, Discount, Customer Satisfaction, and Table Name. The data rows show various delivery ratings, discounts, and satisfaction levels for different orders.

Delivery Rating	Discount	Customer Satisfaction	Table Name
4.20000	0.100000	85 %	Food_Delivery_Orders.csv
4.50000	0.150000	90 %	Food_Delivery_Orders.csv
3.80000	0.050000	78 %	Food_Delivery_Orders.csv
4.70000	0.200000	95 %	Food_Delivery_Orders.csv
3.20000	0.250000	65 %	Food_Delivery_Orders.csv

The screenshot shows the Tableau Desktop interface with the 'Patient_Records' data source selected. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' shelf is set to 'Automatic'. A dialog box titled 'Order Batch' is open, showing a calculated field formula. The formula is: IF CONTAINS([Table Name], "Food_Delivery_Orders 2") THEN "Order Batch 2" ELSEIF CONTAINS([Table Name], "Food_Delivery_Orders") THEN "Order Batch 1" ELSE "Other" END. The dialog box also includes a 'The calculation is valid.' message and 'Apply' and 'OK' buttons.

```
IF CONTAINS([Table Name], "Food_Delivery_Orders 2") THEN "Order Batch 2"
ELSEIF CONTAINS([Table Name], "Food_Delivery_Orders") THEN "Order Batch 1"
ELSE "Other"
END
```

D: Execute Cross-Database Joins

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

indian_start..._funding.csv
List of Orders.csv
Most Strea...olumns.csv
Orders.csv
Patient_Feedback.csv
Patient_Records.csv
Restaurant_Master.csv
startup_funding.csv
store_sales_50_rows.csv
Students - Students.csv
titanic.csv
Treatment_Records.csv

New Union
New Table Extension

Patient_Records

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Deli... — Restaurant...

4 rows

Name Restaurant_Master.csv

How do relationships differ from joins? [Learn more](#)

Food_Delivery_Ord... Operator Restaurant_Master...

Abc Restaurant Cod... = Abc Restaurant Cod...

Add more fields

Performance Options

Restaurant Code (Restaur	Restaurant Name	Cuisine	City	Targ
RES-01	Spice Garden	Indian	Mumbai	
RES-02	Urban Tadka	North Indian	Pune	
RES-03	Food Street	Multi-Cuisine	Mumbai	
RES-04	The Food Hub	Continental	Delhi	

Tableau - Dheeraj 9th

File Data Server Window Help

Connections

Patient_Records
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

indian_start..._funding.csv
List of Orders.csv
Most Strea...olumns.csv
Orders.csv
Patient_Feedback.csv
Patient_Records.csv
Restaurant_Master.csv
startup_funding.csv
store_sales_50_rows.csv
Students - Students.csv
titanic.csv
Treatment_Records.csv

New Union
New Table Extension

Patient_Records

Connection
☒ Live ☐ Extract

Filters
0 | Add

Food_Deli... — Restaurant...

4 rows

Name Restaurant_Master.csv

How do relationships differ from joins? [Learn more](#)

Food_Delivery_Ord... Operator Restaurant_Master...

Abc Restaurant Cod... = Abc Restaurant Cod...

Add more fields

Performance Options

Restaurant Code (Restaur	Restaurant Name	Cuisine	City	Targ
RES-01	Spice Garden	Indian	Mumbai	
RES-02	Urban Tadka	North Indian	Pune	
RES-03	Food Street	Multi-Cuisine	Mumbai	
RES-04	The Food Hub	Continental	Delhi	

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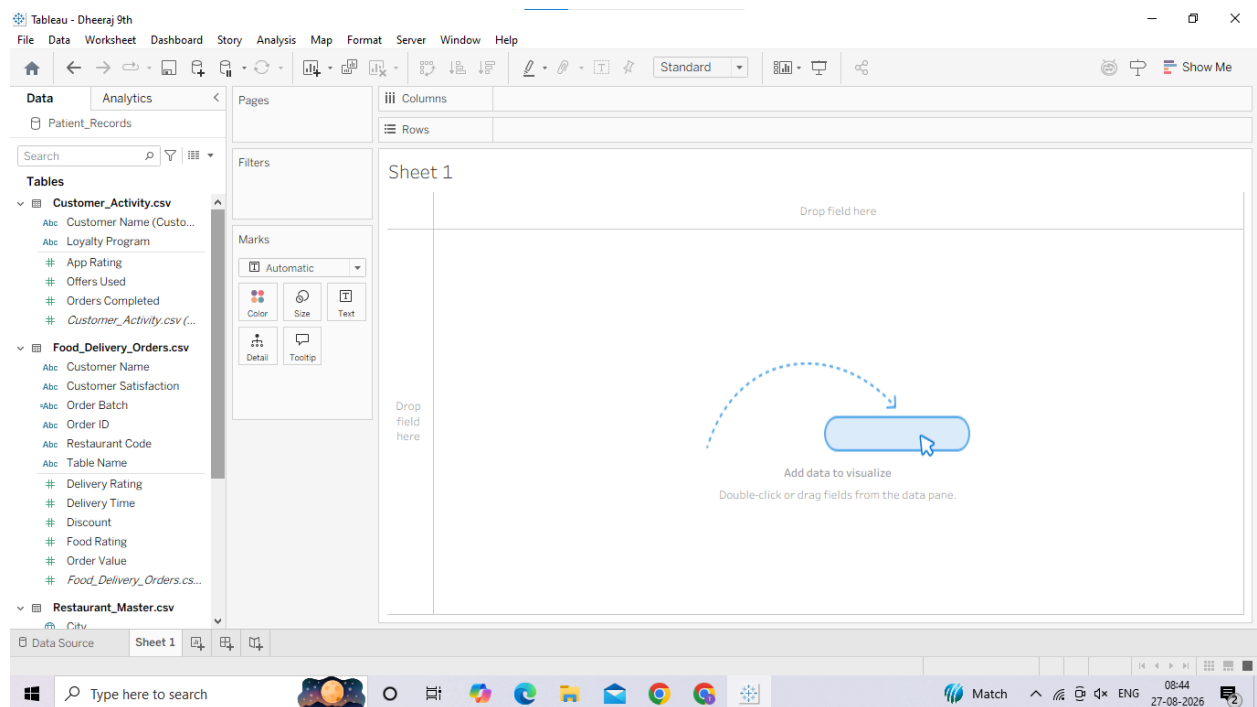
The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Patient_Records' as a text file. Below it, the 'Files' pane shows a list of CSV files including 'Agriculture Production.csv', 'Category_Icons.csv', 'Customer_Activity.csv', 'Data Set- In...ist_2014.csv', 'Department_Master.csv', 'fifa18_clean.csv', 'Food_Delivery_Orders.csv', 'Food_Delive...rders_2.csv', 'INDIA_RETAIL_DATA.csv', 'indian_start...funding.csv', 'List of Orders.csv', and 'Most Strea...olumns.csv'. The main workspace displays a data source connection diagram for 'Patient_Records' showing a join between 'Food_Deli...' and 'Customer_Activity.csv'. Below the diagram, a data preview table is shown with 10 rows. The table has four columns: 'Customer Name (Customi', 'Loyalty Program', 'Orders Completed', and 'Offers Used'. The data rows are: Aarav Sharma (Gold, 12, 5), Priya Verma (Silver, 8, 4), Rohan Gupta (Gold, 10, 3), Sneha Patel (Platinum, 15, 7), and Aritva Rao (Bronze, 5, 2).

Customer Name (Customi	Loyalty Program	Orders Completed	Offers Used
Aarav Sharma	Gold	12	5
Priya Verma	Silver	8	4
Rohan Gupta	Gold	10	3
Sneha Patel	Platinum	15	7
Aritva Rao	Bronze	5	2

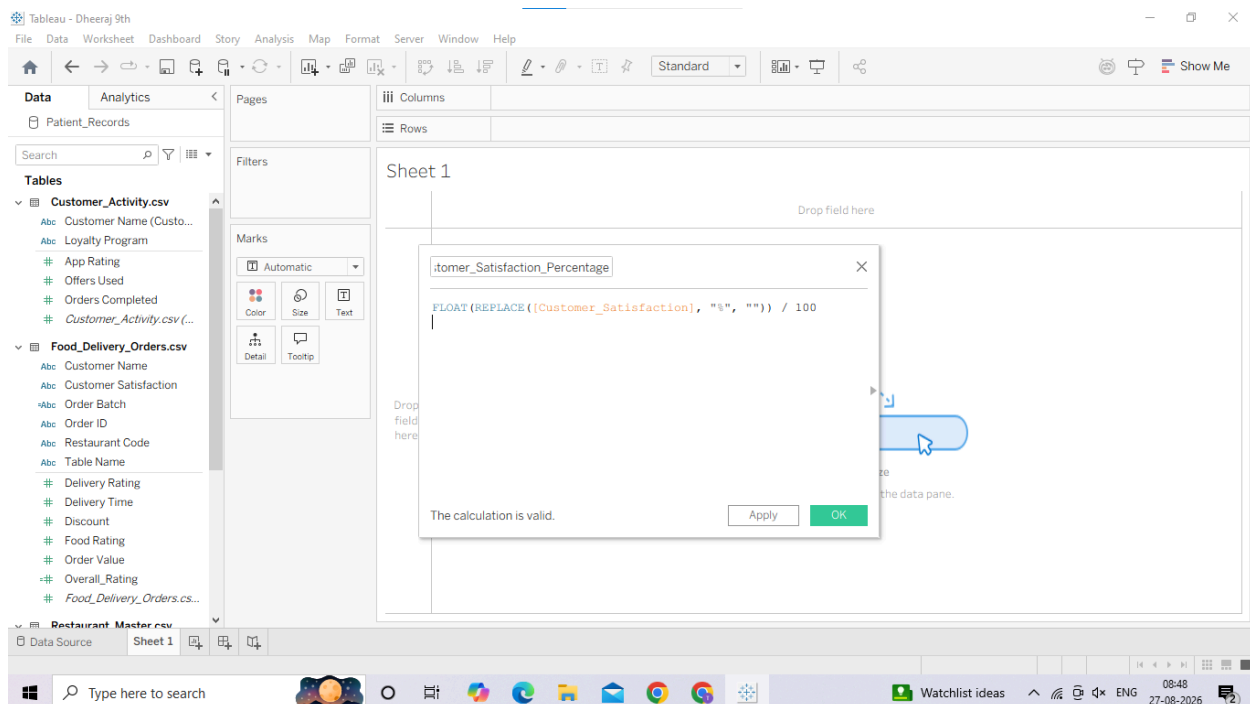
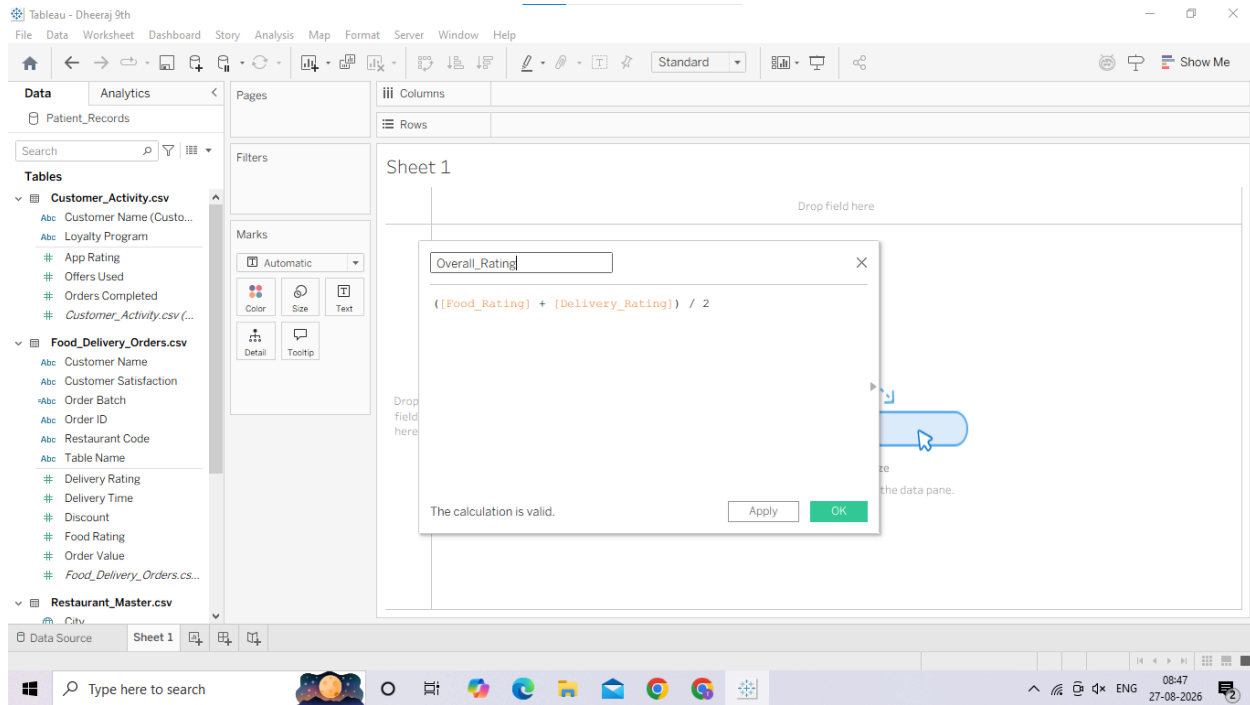
The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Patient_Records' as a text file. Below it, the 'Files' pane shows a list of CSV files including 'Agriculture Production.csv', 'Category_Icons.csv', 'Customer_Activity.csv', 'Data Set- In...ist_2014.csv', 'Department_Master.csv', 'fifa18_clean.csv', 'Food_Delivery_Orders.csv', 'Food_Delive...rders_2.csv', 'INDIA_RETAIL_DATA.csv', 'indian_start...funding.csv', 'List of Orders.csv', and 'Most Strea...olumns.csv'. The main workspace displays a data source connection diagram for 'Patient_Records' showing a join between 'Food_Deli...' and 'Customer_Activity.csv'. Below the diagram, a data preview table is shown with 10 rows. The table has four columns: 'Customer Name (Customi', 'Loyalty Program', 'Orders Completed', and 'Offers Used'. The data rows are: Aarav Sharma (Gold, 12, 5), Priya Verma (Silver, 8, 4), Rohan Gupta (Gold, 10, 3), Sneha Patel (Platinum, 15, 7), and Aritva Rao (Bronze, 5, 2).

Customer Name (Customi	Loyalty Program	Orders Completed	Offers Used
Aarav Sharma	Gold	12	5
Priya Verma	Silver	8	4
Rohan Gupta	Gold	10	3
Sneha Patel	Platinum	15	7
Aritva Rao	Bronze	5	2

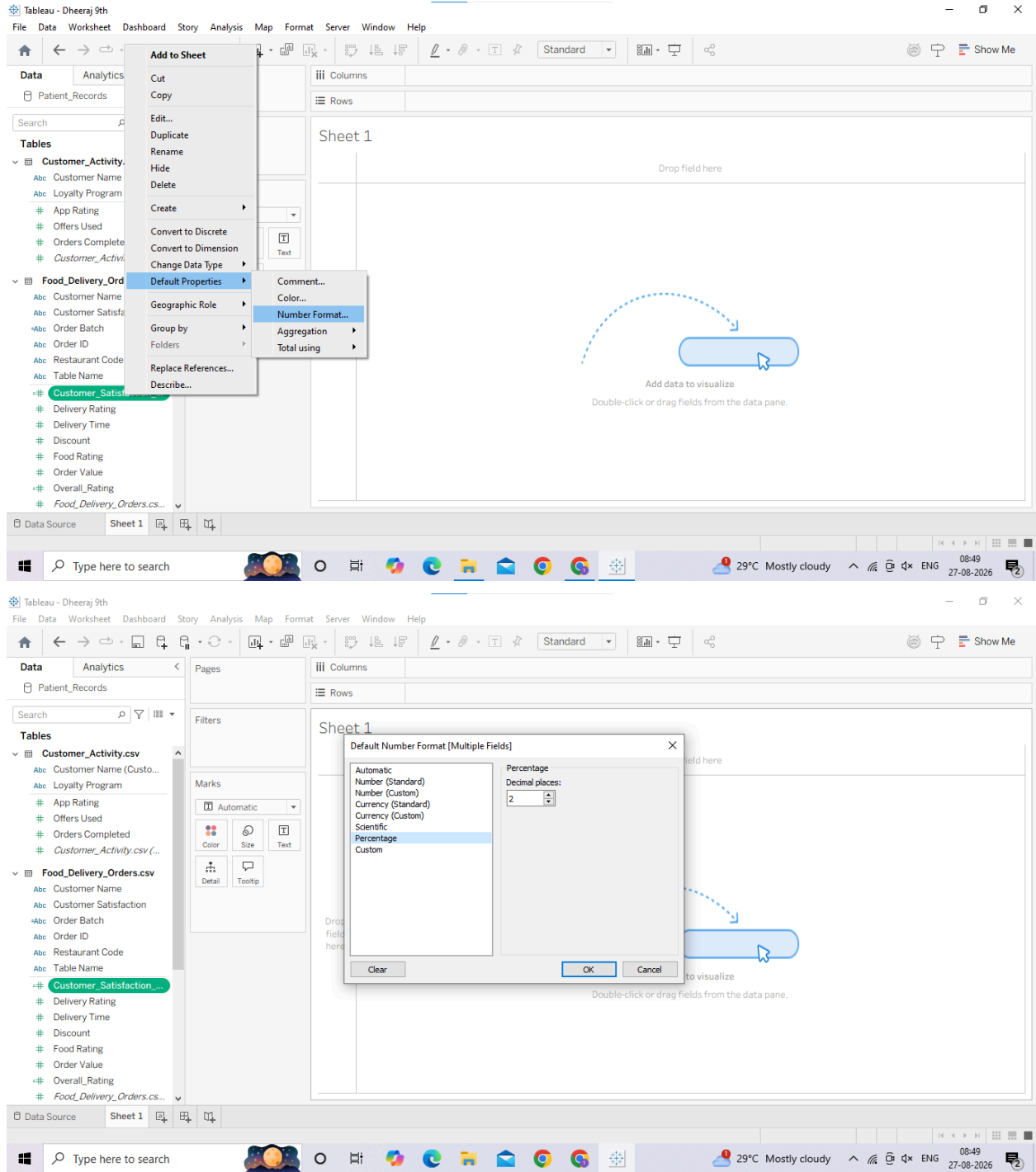
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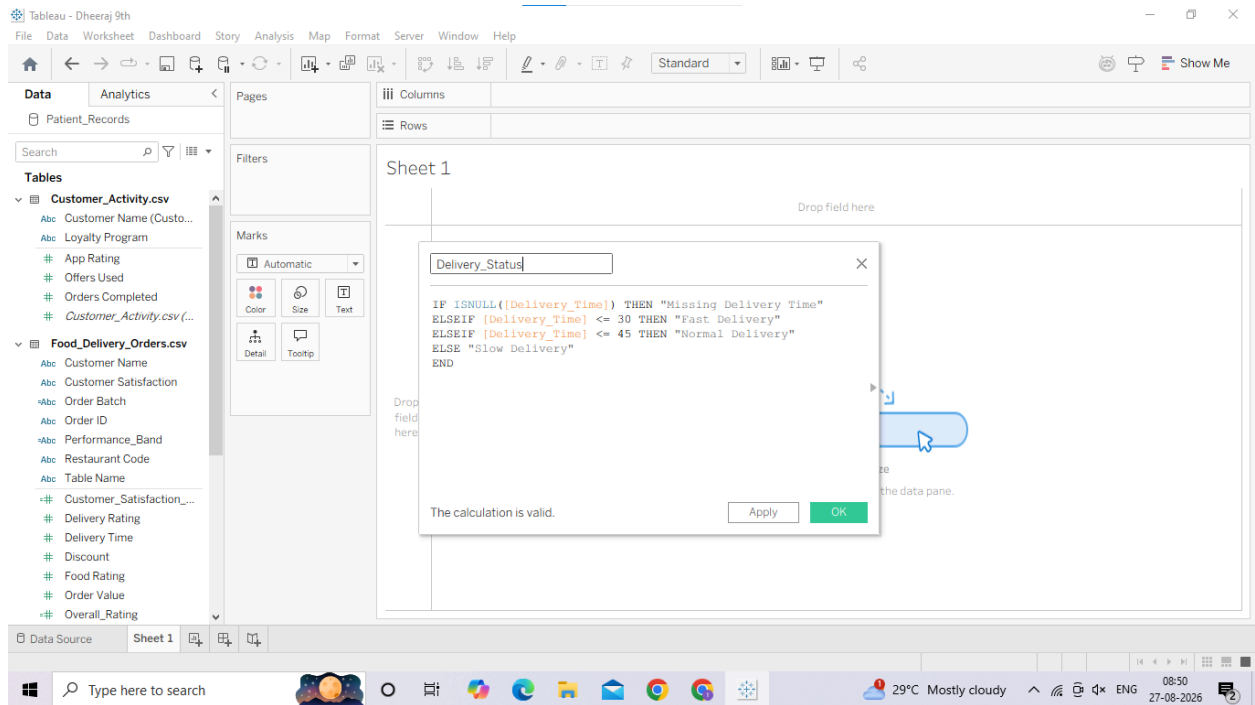
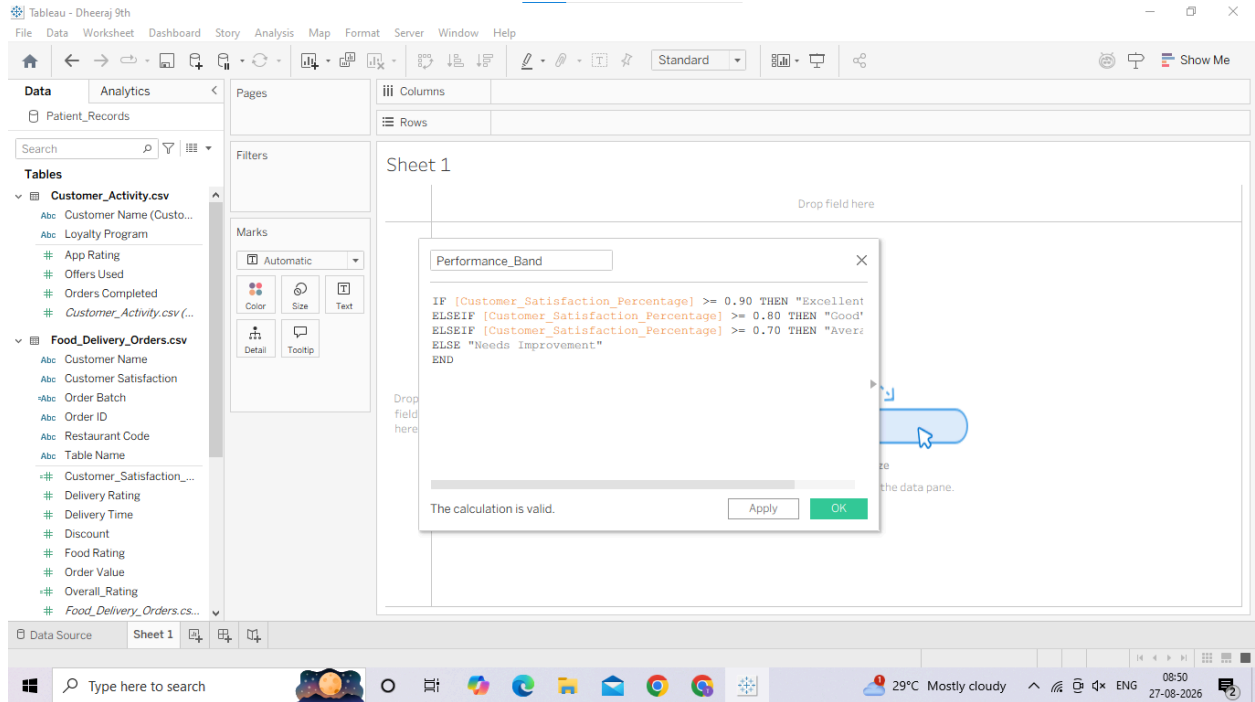
E: Restructure Data for Better Visualization



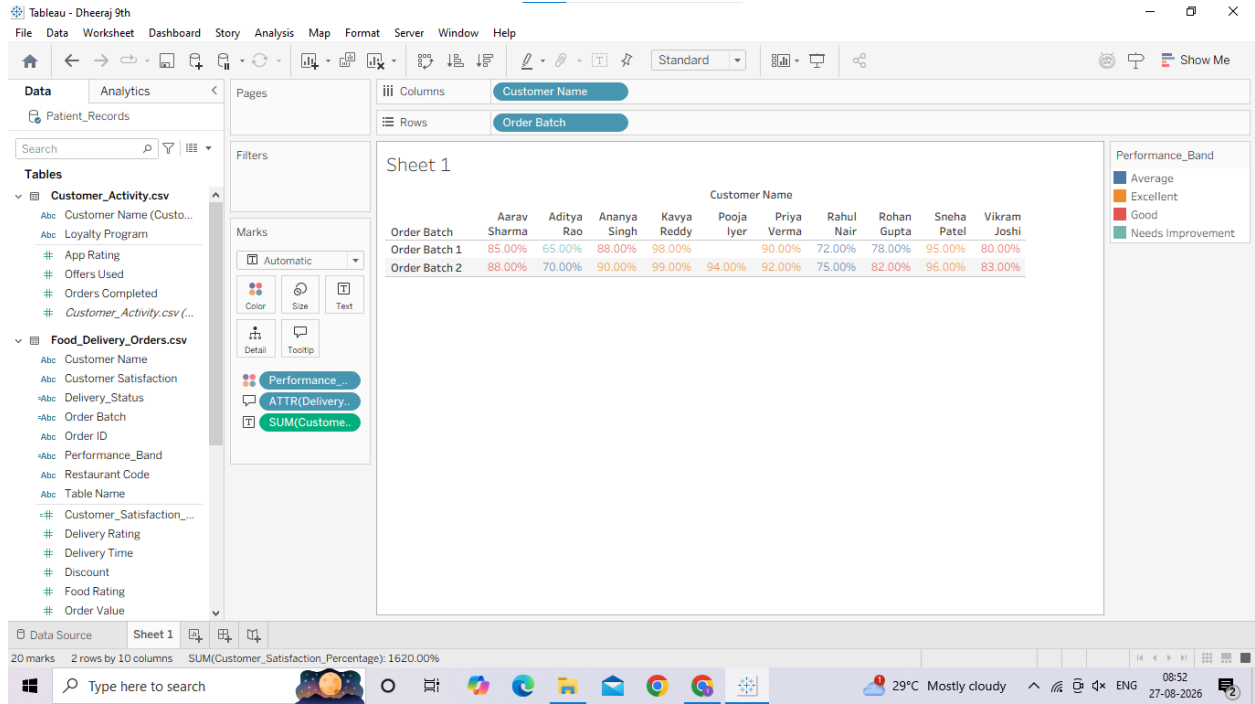
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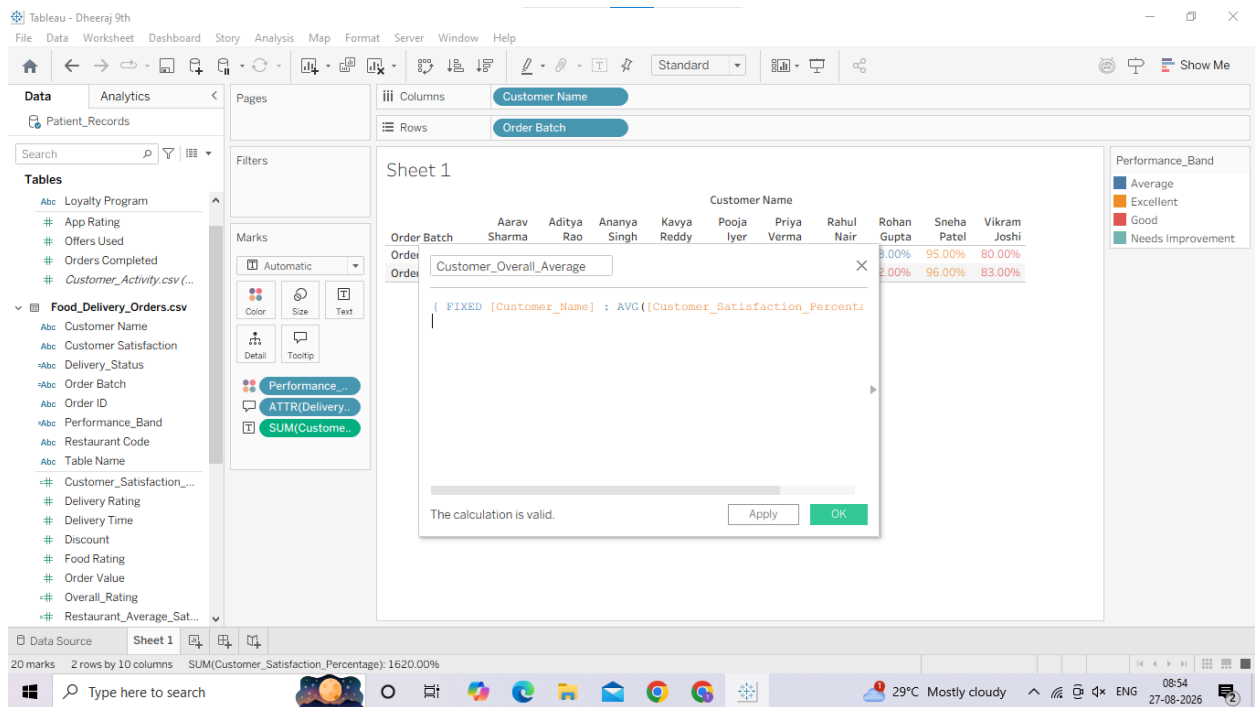
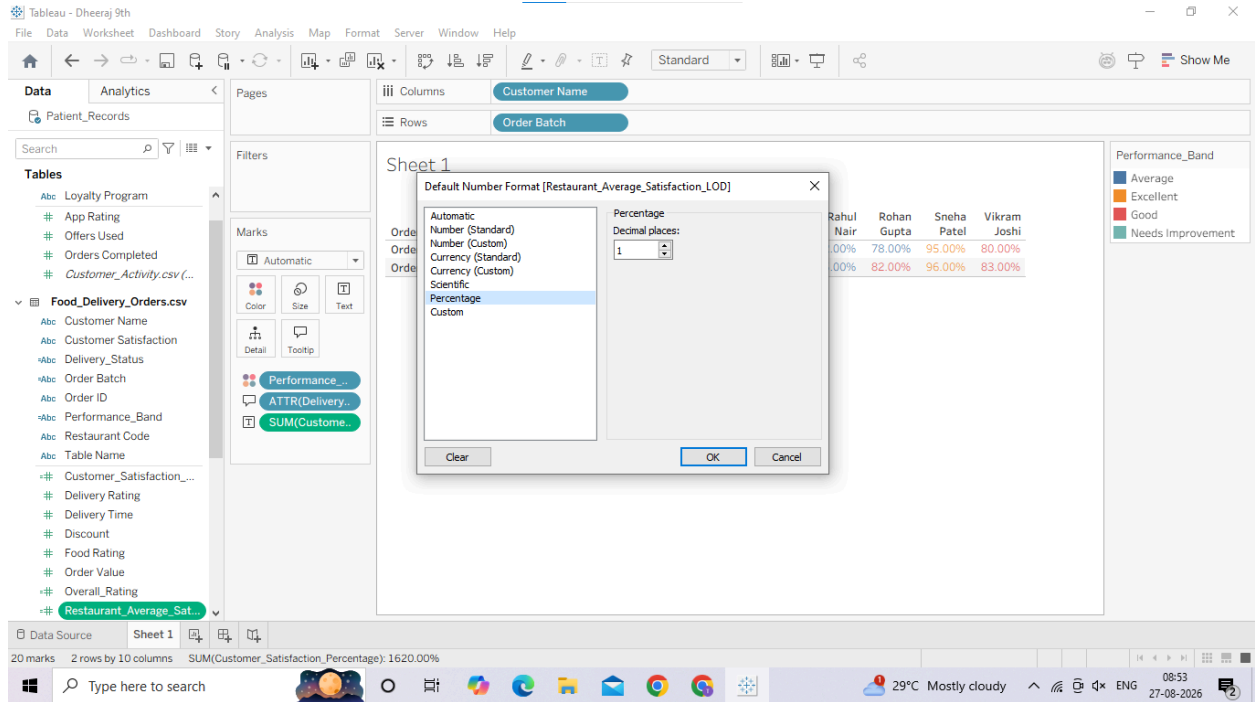
Data Visualization with Power BI and Tableau

F: Handle Different Levels of Detail in Datasets

The screenshot shows the Tableau Desktop interface. The 'Columns' shelf contains 'Customer Name' and the 'Rows' shelf contains 'Order Batch'. A calculation dialog box is open, showing the formula: `{ FIXED [Restaurant_Code] : AVG ([Customer_Satisfaction_Percentage]) }`. The dialog box has 'Apply' and 'OK' buttons. The background view shows a table with columns for 'Order Batch' and 'Customer Name' (Aarav Sharma, Aditya Rao, Ananya Singh, Kavya Reddy, Pooja Iyer, Priya Verma, Rahul Nair, Rohan Gupta, Sneha Patel, Vikram Joshi) and rows for 'Order Batch 1' and 'Order Batch 2'.

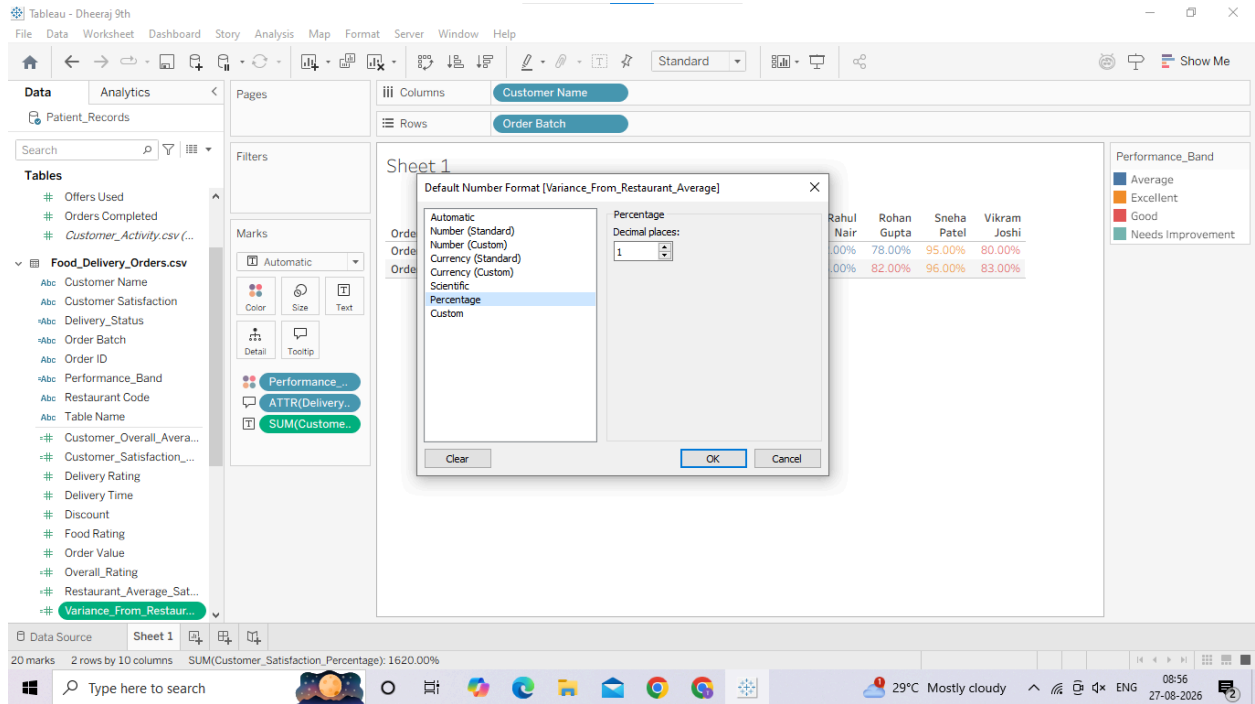
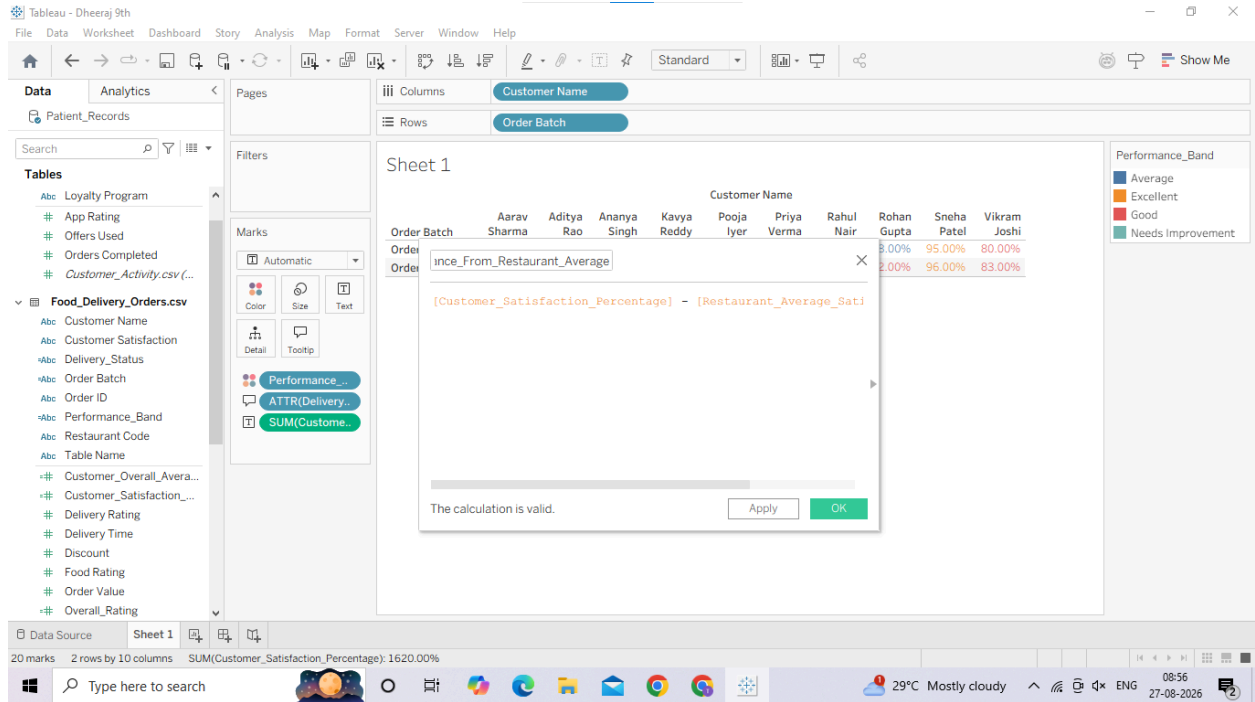
The screenshot shows the Tableau Desktop interface with a context menu open for the 'Restaurant_Average_Sat...' field. The menu options include: Add to Sheet, Show Filter, Cut, Copy, Edit..., Duplicate, Rename, Hide, Delete, Create, Convert to Discrete, Convert to Dimension, Change Data Type, Default Properties, Geographic Role, Group by, Folders, Replace References..., and Describe... The 'Default Properties' option is selected, and a sub-menu is open showing 'Comment...', 'Color...', 'Number Format...', 'Aggregation', and 'Total using'.

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The screenshot displays the Tableau Desktop interface. On the left, the 'Data Source' pane shows a table named 'Customer' with fields like 'Customer Name', 'Customer Satisfaction', and 'Overall_Rating'. The 'Columns' shelf contains 'Customer Name' and the 'Rows' shelf contains 'Order Batch'. The 'Marks' shelf is set to 'Automatic'. A context menu is open over the 'Customer' field, showing options like 'Cut', 'Copy', 'Edit...', 'Duplicate', 'Rename', 'Hide', 'Delete', 'Create', 'Convert to Discrete', 'Convert to Dimension', 'Change Data Type', 'Default Properties', 'Geographic Role', 'Group by', 'Folders', 'Replace References...', and 'Describe...'. The 'Default Properties' sub-menu is open, showing 'Comment...', 'Color...', 'Number Format...', 'Aggregation', and 'Total using'. The 'Number Format...' dialog box is open, showing the 'Default Number Format [Customer_Overall_Average]' window. The 'Automatic' tab is selected, and the 'Percentage' format is chosen. The 'Decimal places' are set to 1. The background view shows a table with columns for 'Customer Name' and 'Order Batch', and rows for 'Order Batch 1' and 'Order Batch 2'. The table is color-coded by 'Performance_Band' (Average, Excellent, Good, Needs Improvement).

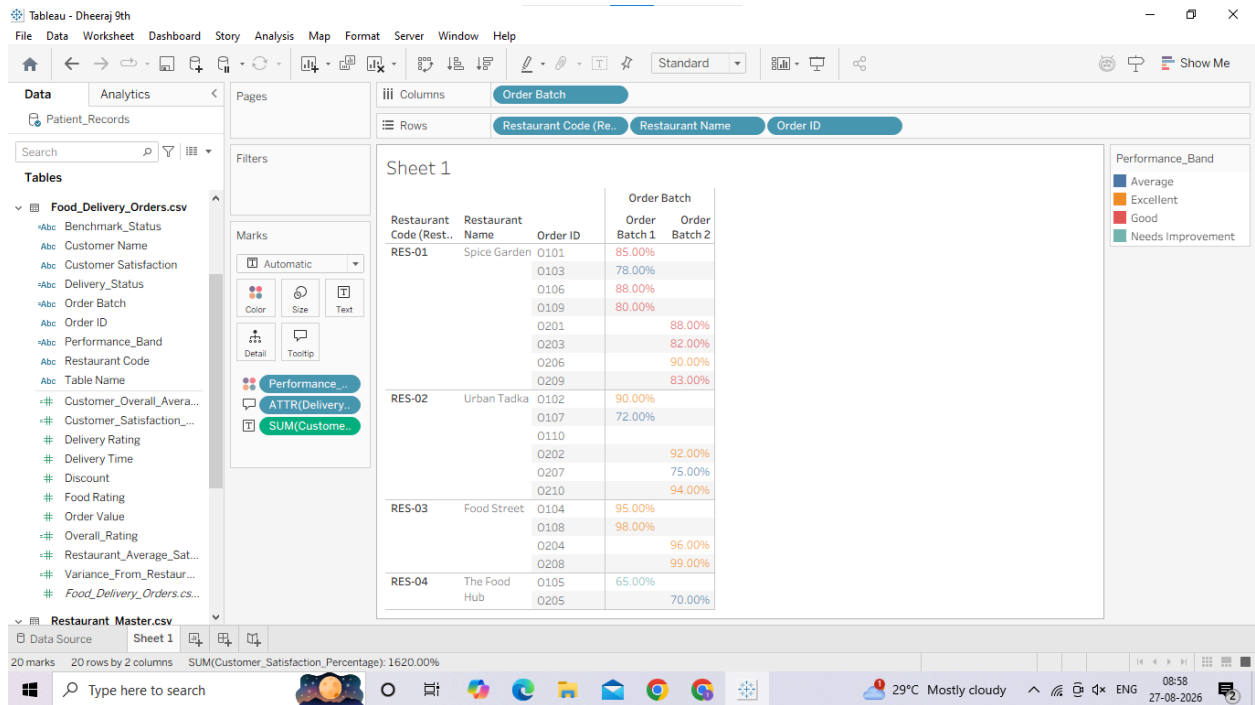
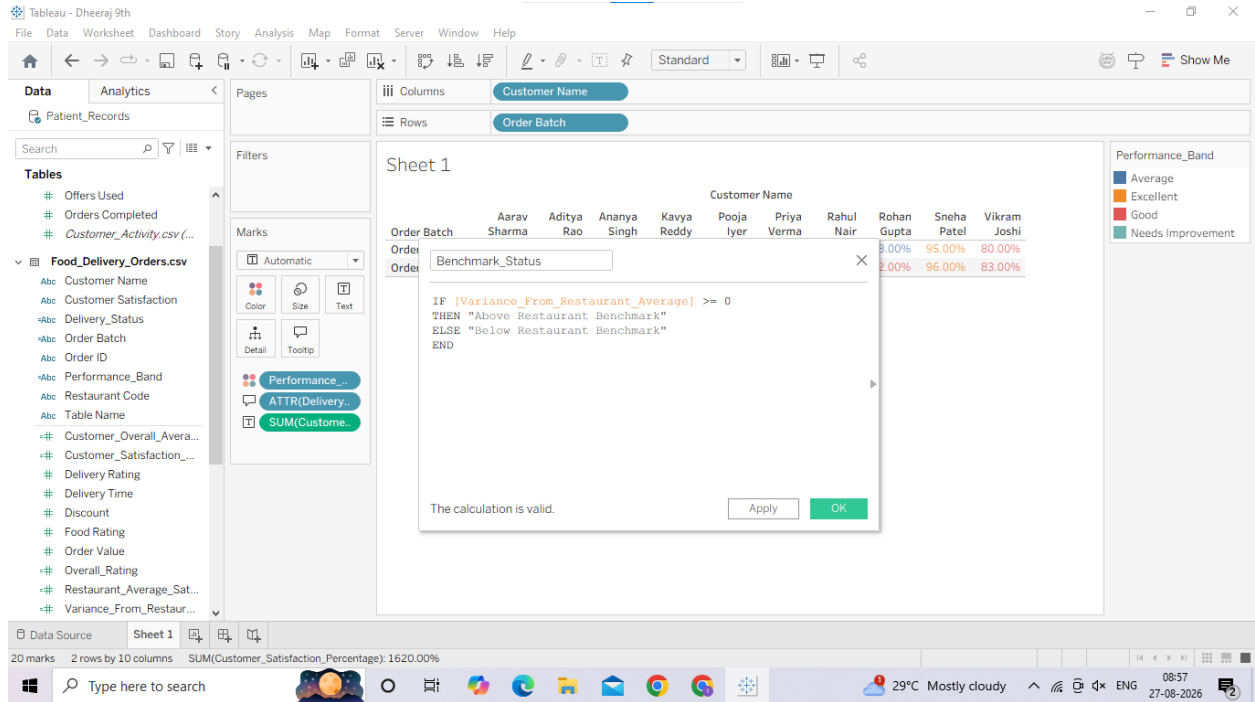
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G: Apply Advanced Data Cleaning Techniques

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Data Analytics

Search

Tables

- Food_Delivery_Orders.csv
 - Benchmark_Status
 - Customer Name
 - Customer Satisfaction
 - Delivery_Status
 - Order Batch
 - Order ID
 - Performance_Band
 - Restaurant Code
 - Table Name
 - Customer_Overall_Avera...
 - Customer_Satisfaction...
 - Delivery Rating
 - Delivery Time
 - Discount
 - Food Rating
 - Order Value
 - Overall_Rating
 - Restaurant_Average_Sat...
 - Variance_From_Restaur...
 - Food_Delivery_Orders.cs...
- Restaurant_Master.csv

Columns: Order Batch

Rows: Restaurant Code (Rest...), Restaurant Name, Order ID

Filters

Marks: Automatic

Performance_Band

- Average
- Excellent
- Good
- Needs Improvement

Sheet 1

Restaurant Code (Rest...), Restaurant Name, Order ID, Order Batch

Customer_Name_Clean

TRIM([Customer_Name])

The calculation is valid.

Apply OK

20 marks 20 rows by 2 columns SUM(Customer_Satisfaction, Percentage): 1620.00%

Type here to search

29°C Mostly cloudy 09:00 27-08-2026

Tableau - Dheeraj 9th

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Data Analytics

Search

Tables

- Food_Delivery_Orders.csv
 - Benchmark_Status
 - Customer Name
 - Customer Satisfaction
 - Customer_Name_Clean
 - Delivery_Status
 - Order Batch
 - Order ID
 - Performance_Band
 - Restaurant Code
 - Table Name
 - Customer_Overall_Avera...
 - Customer_Satisfaction...
 - Delivery Rating
 - Delivery Time
 - Discount
 - Food Rating
 - Order Value
 - Overall_Rating
 - Restaurant_Average_Sat...
 - Variance_From_Restaur...
 - Food_Delivery_Orders.cs...
- Restaurant_Master.csv

Columns: Order Batch

Rows: Restaurant Code (Rest...), Restaurant Name, Order ID

Filters

Marks: Automatic

Performance_Band

- Average
- Excellent
- Good
- Needs Improvement

Sheet 1

Restaurant Code (Rest...), Restaurant Name, Order ID, Order Batch

First_Name

TRIM(SPLIT([Customer_Name], " ", 1))

The calculation is valid.

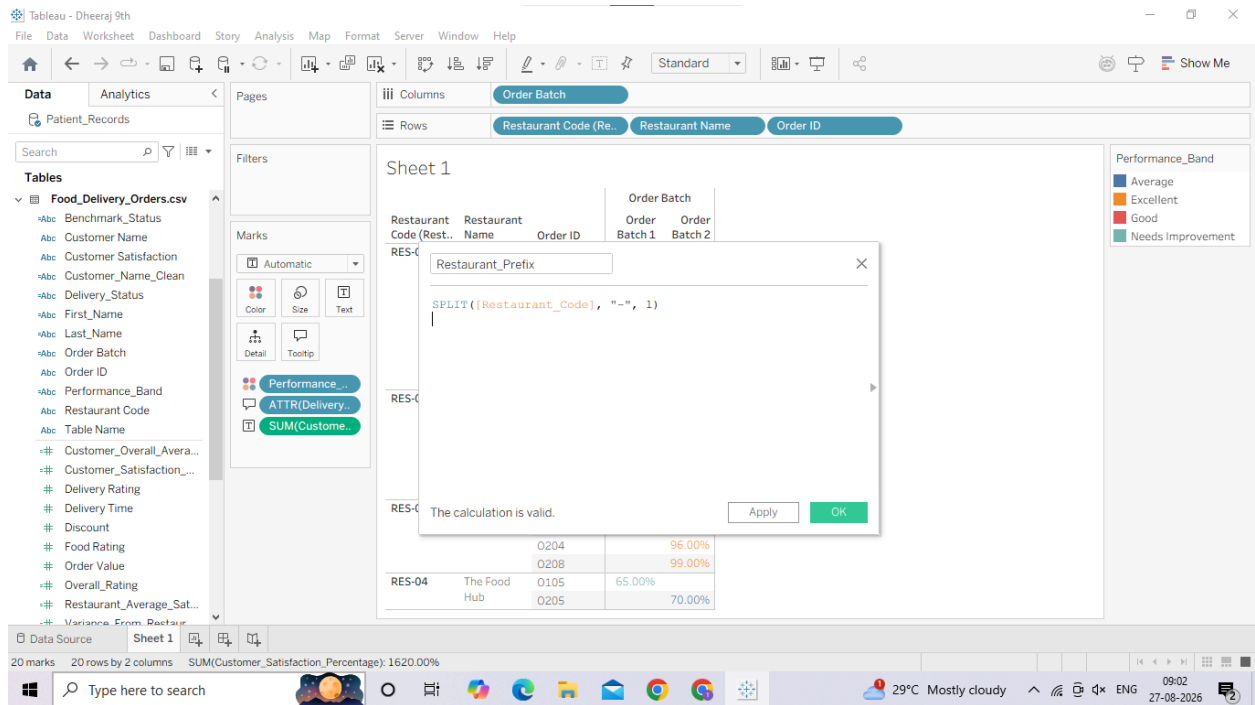
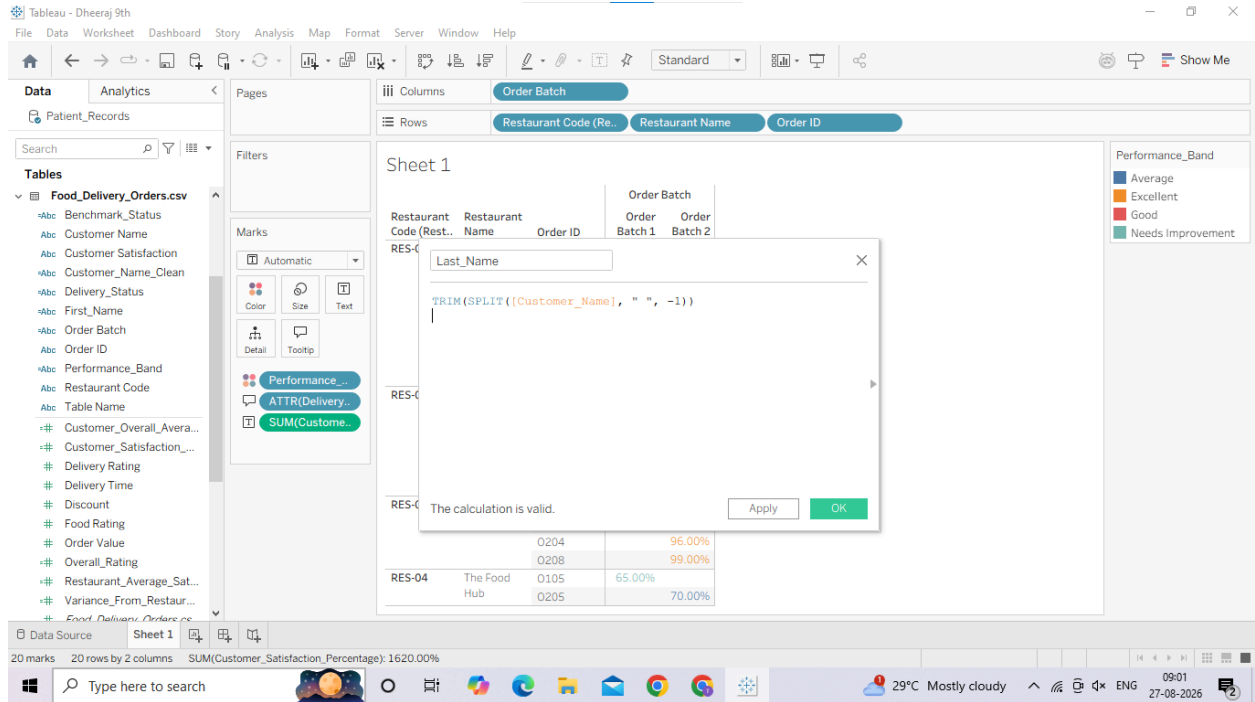
Apply OK

20 marks 20 rows by 2 columns SUM(Customer_Satisfaction, Percentage): 1620.00%

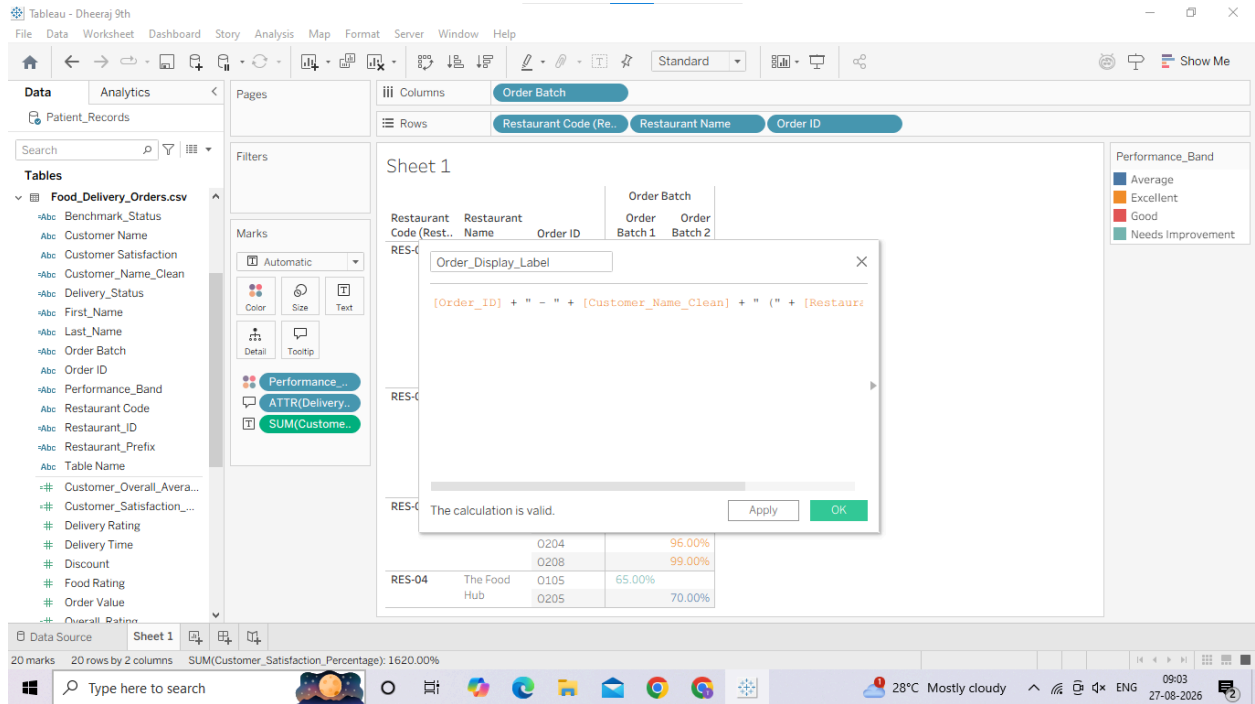
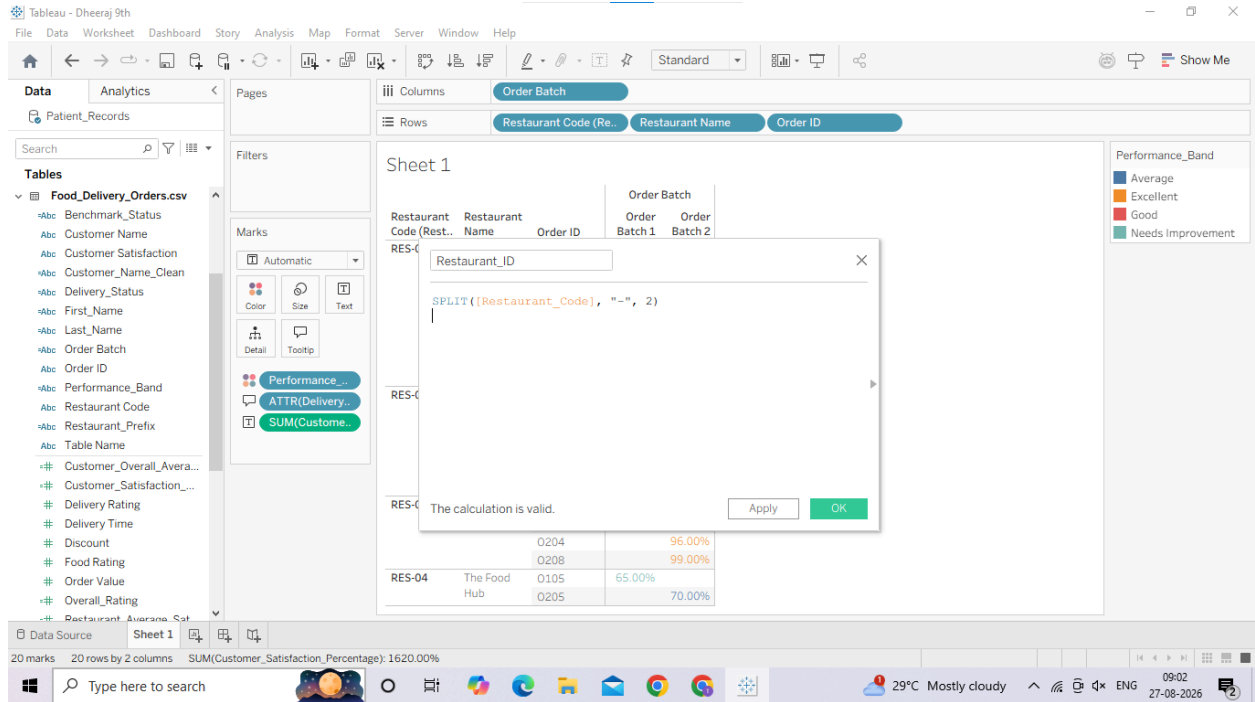
Type here to search

29°C Mostly cloudy 09:01 27-08-2026

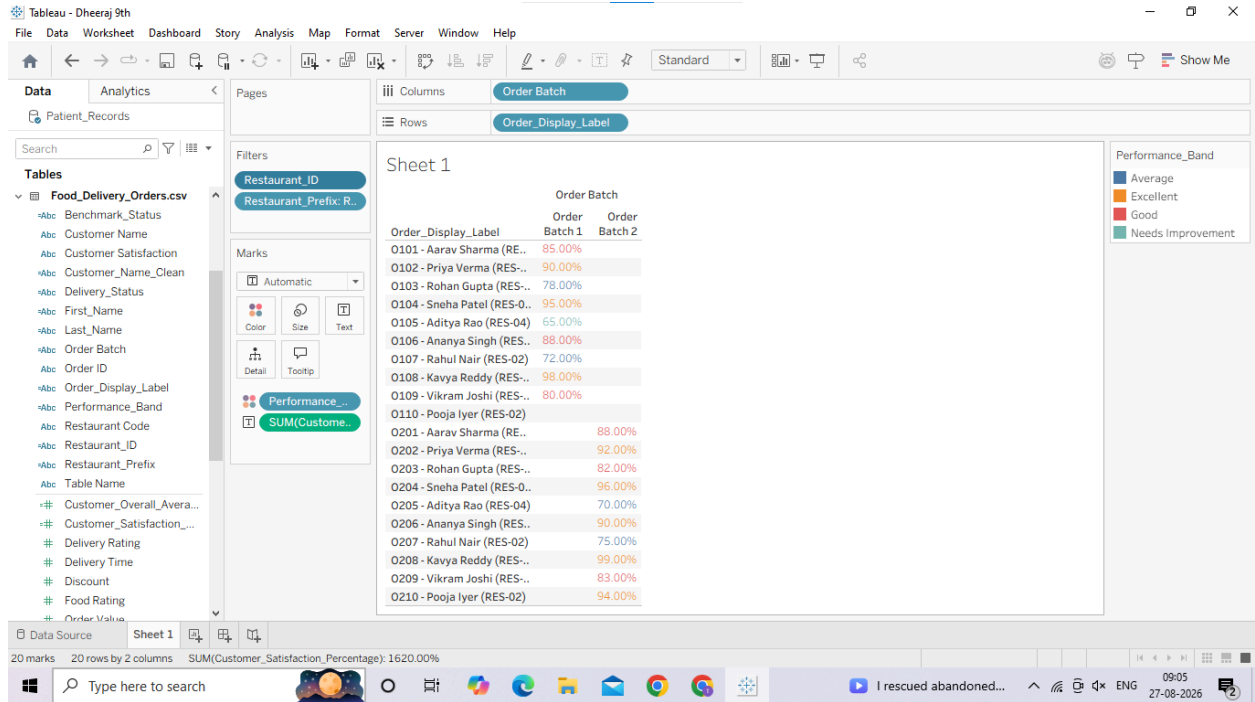
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